LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Descend forces in a descend glide range.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Descend forces in a descend glide range.
5. Specific Objectives	11.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight performance charts; Instructor supervision required for chart interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aircraft in flight?
(b) How does the weight vector change when an aircraft enters a glide?
(c) Define the term 'glide range' in the context of helicopter theory.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of descend forces • Weight vector orientation in glide • Lift and drag relationship • Thrust absence in glide • Glide angle determination	L & Ds	PPT, IB	15 min
(b)	• Calculation of best glide speed • Effect of weight on glide ratio • Wind correction in descent • Energy management during glide • Stall speed in glide configuration	L & Ds	PPT, PDF	20 min
(c)	• Recovery techniques from glide • Emergency descent procedures • Fuel conservation in glide • Landing zone selection criteria • Final approach alignment	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the absence of engine thrust alters the force equilibrium in a helicopter glide.
(b) Describe the relationship between the glide angle and the lift-to-drag ratio.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of descend forces in a glide range apply to the initial approach phase described in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Turn forces in a turn, rate of turn steady level coordinated turn and control in a turn.	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min